

## LIVERPOOL JOHN LENNON, United Kingdom

WMO: 033233

Lat: 53.334N

Lon: 2.850W

Elev: 24

StdP: 101.03

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |            |             |            |            | Coldest Month WS/MCDB |            |             |            | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|------------|-------------|------------|------------|-----------------------|------------|-------------|------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |            | 99%         |            |            | 0.4%                  |            | 1%          |            |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB       | DP          | HR         | MCDB       | WS                    | MCDB       | WS          | MCDB       | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-2.1 | (c)<br>-1.0 | (d)<br>-5.8                   | (e)<br>2.3 | (f)<br>0.4 | (g)<br>-4.2 | (h)<br>2.7 | (i)<br>1.2 | (j)<br>16.5           | (k)<br>8.4 | (l)<br>14.6 | (m)<br>8.4 | (n)<br>2.9            | (o)<br>100 | (p)<br>0.624 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 7.0                    | 25.1            | 17.9 | 23.1 | 17.0 | 21.8 | 16.6 | 18.7                | 23.1 | 17.9 | 21.9 | 17.1 | 20.5 | 4.2                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.1                            | 12.2 | 20.4 | 16.2 | 11.5 | 19.6 | 15.8 | 11.2 | 19.2 | 53.2          | 23.2 | 50.5 | 22.0 | 48.2 | 20.7 | 23.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
| (a)               | (b)  | (c) | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| 12.7              | 11.1 | 9.8 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
|                   |      |     | -5.0                       | 28.4 | 2.2                | 2.1 | -6.6                                               | 29.9 | -7.9       | 31.1 | -9.2       | 32.3 | -10.8      | 33.8 |
|                   |      |     | -5.5                       | 20.8 | 2.2                | 1.3 | -7.1                                               | 21.7 | -8.4       | 22.5 | -9.6       | 23.3 | -11.2      | 24.2 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 10.7   | 5.3   | 5.6   | 7.2   | 9.3   | 12.3  | 15.1  | 17.0  | 16.8  | 14.5  | 11.5  | 8.1   | 5.7  |
|                                                                                  | DBStd       | 4.99   | 2.96  | 2.88  | 2.68  | 2.72  | 2.73  | 2.41  | 2.34  | 2.05  | 2.11  | 2.57  | 2.84  | 3.40 |
|                                                                                  | HDD10.0     | 634    | 146   | 124   | 94    | 43    | 8     | 0     | 0     | 0     | 0     | 14    | 69    | 136  |
|                                                                                  | HDD18.3     | 2807   | 403   | 356   | 347   | 269   | 188   | 103   | 57    | 58    | 116   | 212   | 308   | 391  |
|                                                                                  | CDD10.0     | 899    | 1     | 2     | 6     | 23    | 79    | 152   | 217   | 209   | 136   | 61    | 11    | 3    |
|                                                                                  | CDD18.3     | 31     | 0     | 0     | 0     | 0     | 1     | 4     | 15    | 9     | 2     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 165    | 0     | 0     | 0     | 0     | 4     | 31    | 85    | 41    | 3     | 1     | 0     | 0    |
|                                                                                  | CDH26.7     | 26     | 0     | 0     | 0     | 0     | 0     | 4     | 14    | 7     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 4.7    | 5.3   | 5.2   | 5.1   | 4.7   | 4.6   | 4.4   | 4.3   | 4.3   | 4.3   | 4.7   | 4.7   | 4.9  |
| Precipitation                                                                    | PrecAvg     | 801    | 70    | 53    | 48    | 54    | 59    | 63    | 66    | 74    | 69    | 84    | 81    | 82   |
|                                                                                  | PrecMax     | 1088   | 133   | 100   | 104   | 111   | 110   | 179   | 133   | 171   | 131   | 171   | 174   | 150  |
|                                                                                  | PrecMin     | 578    | 9     | 7     | 9     | 13    | 10    | 19    | 17    | 19    | 9     | 19    | 41    | 27   |
|                                                                                  | PrecStd     | 120    | 32    | 24    | 24    | 28    | 25    | 38    | 28    | 31    | 31    | 35    | 30    | 32   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 13.2  | 13.9  | 16.0  | 20.2  | 23.9  | 26.9  | 28.8  | 27.2  | 23.8  | 19.0  | 15.8  | 14.0 |
|                                                                                  |             | MCWB   | 10.9  | 11.2  | 11.4  | 13.8  | 16.6  | 17.8  | 18.9  | 19.0  | 17.6  | 15.9  | 13.8  | 12.0 |
|                                                                                  | 2%          | DB     | 12.0  | 12.1  | 13.9  | 17.2  | 21.2  | 23.2  | 25.8  | 23.9  | 21.1  | 17.2  | 14.1  | 12.9 |
|                                                                                  |             | MCWB   | 9.8   | 9.8   | 10.5  | 12.3  | 15.4  | 16.9  | 17.9  | 17.8  | 16.5  | 14.8  | 12.3  | 11.1 |
|                                                                                  | 5%          | DB     | 10.9  | 11.0  | 12.2  | 15.2  | 19.0  | 21.1  | 23.2  | 21.9  | 19.2  | 16.2  | 13.1  | 11.8 |
|                                                                                  |             | MCWB   | 8.9   | 8.8   | 9.5   | 11.1  | 14.0  | 15.9  | 17.1  | 17.2  | 15.4  | 13.9  | 11.2  | 10.1 |
|                                                                                  | 10%         | DB     | 9.8   | 10.0  | 11.2  | 13.9  | 17.0  | 19.8  | 21.2  | 20.2  | 18.1  | 15.2  | 12.1  | 10.2 |
|                                                                                  |             | MCWB   | 8.3   | 8.2   | 8.6   | 10.4  | 13.0  | 15.0  | 16.5  | 16.2  | 14.6  | 13.0  | 10.3  | 8.7  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 11.4  | 11.5  | 12.1  | 14.2  | 17.6  | 19.4  | 20.1  | 19.6  | 18.5  | 16.5  | 14.0  | 12.5 |
|                                                                                  |             | MCDB   | 13.0  | 13.5  | 14.8  | 18.8  | 22.8  | 24.2  | 25.4  | 24.1  | 22.0  | 18.3  | 15.4  | 13.7 |
|                                                                                  | 2%          | WB     | 10.2  | 10.3  | 11.0  | 12.8  | 15.8  | 17.6  | 18.8  | 18.6  | 17.3  | 15.2  | 12.8  | 11.4 |
|                                                                                  |             | MCDB   | 11.5  | 11.9  | 13.3  | 16.4  | 20.2  | 21.7  | 23.8  | 22.5  | 20.1  | 16.9  | 14.2  | 12.8 |
|                                                                                  | 5%          | WB     | 9.2   | 9.2   | 10.1  | 11.6  | 14.6  | 16.4  | 17.8  | 17.7  | 16.1  | 14.2  | 11.6  | 10.3 |
|                                                                                  |             | MCDB   | 10.4  | 10.7  | 12.1  | 14.8  | 18.3  | 20.1  | 22.2  | 21.1  | 18.5  | 16.0  | 13.0  | 11.5 |
|                                                                                  | 10%         | WB     | 8.2   | 8.3   | 9.2   | 10.6  | 13.3  | 15.4  | 16.9  | 17.0  | 15.2  | 13.2  | 10.7  | 9.1  |
|                                                                                  |             | MCDB   | 9.5   | 9.8   | 11.1  | 13.4  | 16.5  | 18.9  | 20.6  | 19.7  | 17.6  | 15.0  | 12.0  | 10.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.4   | 5.1   | 6.0   | 6.9   | 7.3   | 7.1   | 7.0   | 6.5   | 6.3   | 5.6   | 4.6   | 4.5  |
|                                                                                  |             | MCDBR  | 5.4   | 5.8   | 7.3   | 9.4   | 10.8  | 10.5  | 10.7  | 9.2   | 8.3   | 5.9   | 5.1   | 5.3  |
|                                                                                  |             | MCWBR  | 5.0   | 4.6   | 5.1   | 5.7   | 6.2   | 5.6   | 5.0   | 4.8   | 4.9   | 4.5   | 4.5   | 4.8  |
|                                                                                  | 5% WB       | MCDBR  | 5.2   | 5.6   | 6.7   | 8.5   | 9.6   | 9.0   | 9.2   | 7.8   | 7.1   | 5.7   | 5.0   | 5.2  |
|                                                                                  |             | MCWBR  | 5.0   | 4.8   | 5.0   | 5.5   | 5.9   | 5.4   | 4.9   | 4.6   | 4.9   | 4.7   | 4.5   | 5.0  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.326  | 0.341 | 0.371 | 0.394 | 0.398 | 0.395 | 0.395 | 0.381 | 0.368 | 0.350 | 0.326 | 0.316 |      |
|                                                                                  | taud        | 2.321  | 2.375 | 2.300 | 2.252 | 2.285 | 2.323 | 2.328 | 2.383 | 2.420 | 2.432 | 2.385 | 2.297 |      |
|                                                                                  | Ebn at Noon | 636    | 757   | 806   | 834   | 850   | 856   | 848   | 841   | 805   | 737   | 637   | 568   |      |
|                                                                                  | Edh at Noon | 61     | 78    | 102   | 121   | 123   | 120   | 118   | 105   | 90    | 73    | 56    | 52    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.64   | 1.32  | 2.48  | 3.83  | 4.86  | 5.09  | 4.81  | 3.95  | 2.84  | 1.58  | 0.78  | 0.49  |      |
|                                                                                  | RadStd      | 0.06   | 0.14  | 0.21  | 0.35  | 0.34  | 0.47  | 0.48  | 0.26  | 0.21  | 0.13  | 0.08  | 0.05  |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            | N/A   | +0.60   | +0.98  | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (68 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page